

Vishva Arasan

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EDUCATION

University of California Santa Barbara

Santa Barbara, CA

Bachelor of Science in Computer Science, Accelerated Track

June 2027

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, HTML/CSS

Frameworks: Django, Django REST Framework, Flask, Node.js

Tools & Testing: KLEE, DeepSeek V3.2, Git, Postman, PostgreSQL, SQLite, Vim, VS Code, AWS Cloud Practitioner

Core Concepts: LLM Inference, Symbolic Execution, Fuzzing, Static Analysis, REST APIs, ORM

RESEARCH EXPERIENCE

Undergraduate AI Security Research

Sep 2025 – Present

UCSB Verification Lab (Advisor: Prof. Bultan)

Santa Barbara, CA

- DARPA HARDEN Challenge: Engineering STASE (Static Analysis Guided Symbolic Execution), a custom toolchain designed to detect Weird Machine (WMI) vulnerabilities within the Linux Kernel.
- Developing an automated pipeline using **DeepSeek V3.2** to generate C drivers for **KLEE**, automating **symbolic execution** for vulnerability detection.
- Designed an LLM-assisted vulnerability discovery pipeline integrating static analysis triage with **symbolic execution**, successfully accelerating execution speed by **10x** and reducing API consumption by **90%**.
- Engineered automated bug chaining and deep path exploration tools capable of identifying **20-40%** of theoretically chainable vulnerabilities across massive codebases including TCPDump and SQLite.
- Learning comprehensive evaluation frameworks (ExploitGym, ExploitBench) to benchmark LLM code reasoning and SMT solver capabilities in support of the **DARPA HARDEN** challenge.

PROJECTS

KOS Operating System Kernel | *C, k-threads, Semaphores*

Mar 2026

- Engineered an **operating system** from scratch in C implementing 15+ core system calls including process management (fork, execve, wait, exit) and file descriptor handling (dup, dup2, close).
- Architected a thread-safe, atomic inter-process communication (**IPC**) pipe system utilizing circular buffers and strict semaphore synchronization (**mutexes**) to guarantee data integrity across multiple processes.
- Managed process control blocks (**PCBS**) and file descriptor tables, utilizing strict reference counting for readers and writers to prevent memory leaks and ensure safe process blocking/unblocking.

Global Countries Management API | *Python, Django, REST Framework*

Dec 2025

- Developed a **RESTful API** implementing full **CRUD operations** and custom search logic via **Django QuerySets**.
- Standardized **JSON error handling** and data integrity through **ModelSerializers** and input validation logic.

PROFESSIONAL EXPERIENCE

Junior Backend Engineer

June 2026 – Sep 2026

Sesi Technologies

Remote

- Engineered a public-facing **RESTful API** using **Python** and **Django REST Framework** to expose verifiable digital identity records for an agricultural network of 10,000+ users and 20,000+ mapped hectares.
- Architected an end-to-end API usage metering and billing system utilizing **Django ORM** to track client endpoints automatically, enforce rate limits, and apply configurable pricing tiers.
- Formulated response schemas and **API key authentication** protocols that securely share compliance data while strictly protecting sensitive personal information from unauthorized third-party access.

Software Engineering Intern

April 2023 – July 2023

CUBUS Solutions Inc. (Merged with Tyfone)

Livermore, CA

- Supported the development of digital banking services using **SQL databases** and **JavaScript APIs**.